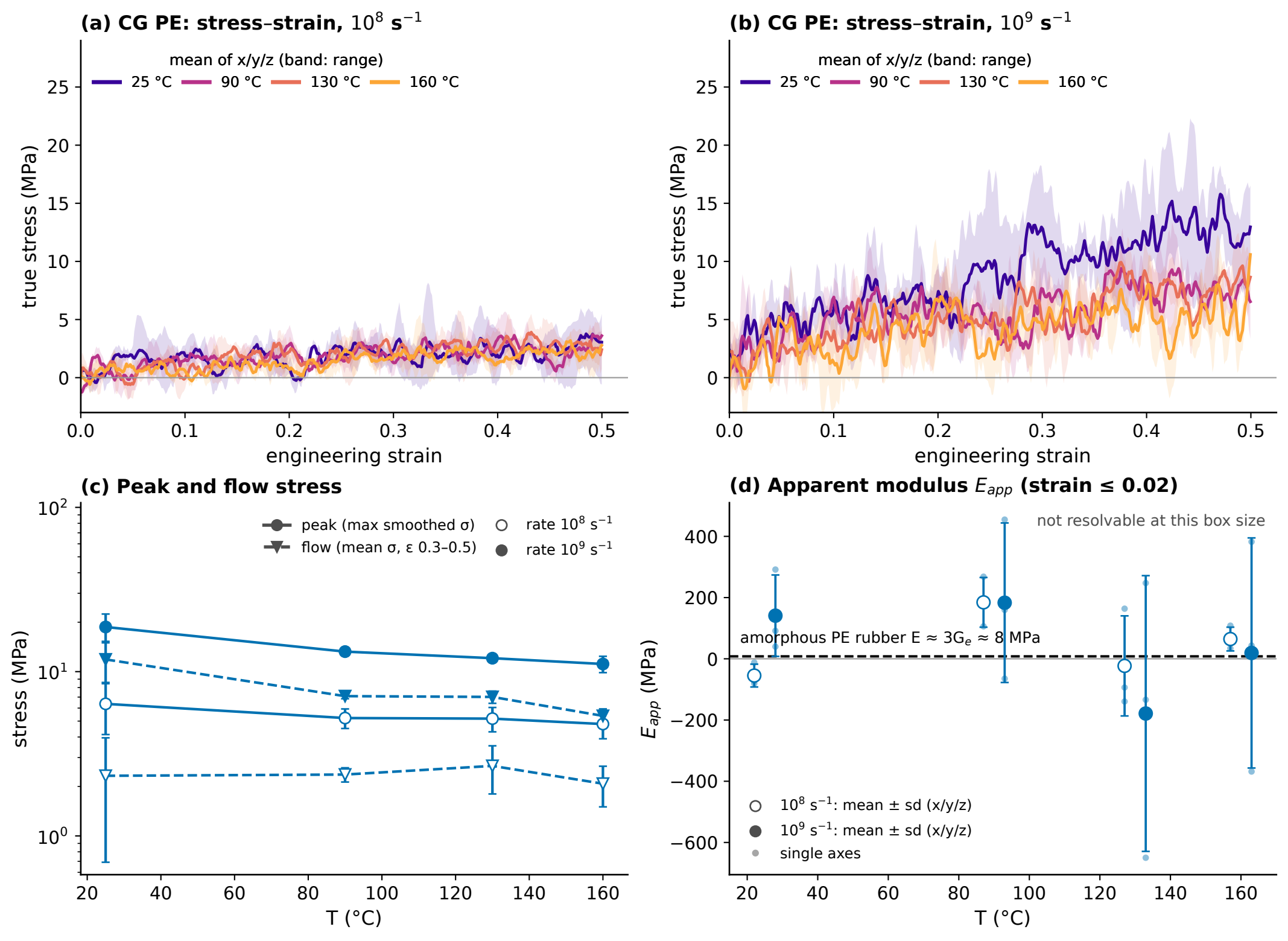

(a) CG PE: stress–strain, $10^8$ s$^{-1}$
mean of x/y/z (band: range)
25 °C
90 °C
130 °C
160 °C
true stress (MPa)
engineering strain
(b) CG PE: stress–strain, $10^9$ s$^{-1}$
mean of x/y/z (band: range)
25 °C
90 °C
130 °C
160 °C
true stress (MPa)
engineering strain
(c) Peak and flow stress
peak (max smoothed σ)
flow (mean σ, ε 0.3–0.5)
rate $10^8$ s$^{-1}$
rate $10^9$ s$^{-1}$
stress (MPa)
T (°C)
(d) Apparent modulus $E_{app}$ (strain ≤ 0.02)
not resolvable at this box size
amorphous PE rubber E ≈ 3G$_e$ ≈ 8 MPa
$10^8$ s$^{-1}$: mean ± sd (x/y/z)
$10^9$ s$^{-1}$: mean ± sd (x/y/z)
single axes
$E_{app}$ (MPa)
T (°C)